

CONJURA · OPEN PROBLEM

Randomness Extraction from Unpredictable Random-Oracle Sources

A Leftover-Hash-Lemma Conjecture, Secret Seed

Status. Statement: human-written, not yet formalized.

Category. Information-Theoretic Cryptography.

Setting. Fix nonempty finite sets $\mathcal{K}$ (seeds), $\mathcal{D}$ (inputs), $\mathcal{R}$ (outputs), and write $K := |\mathcal{K}|$, $D := |\mathcal{D}|$, $R := |\mathcal{R}|$. Let $\text{Fun}(\mathcal{K} \times \mathcal{D}, \mathcal{R})$ be the set of all functions $\mathcal{K} \times \mathcal{D} \rightarrow \mathcal{R}$, let $\text{SD}(\cdot, \cdot)$ denote statistical distance and $U_{\mathcal{R}}$ the uniform distribution on $\mathcal{R}$. The source S , the predictor P and the distinguisher D of Figure 1 are computationally unbounded and receive the *entire* function table of H as an explicit input. The seed sd is sampled after S has terminated, selects the row of H that produces the real challenge, and is *never* given to D : that is what makes this the *secret-seed* game.

Game Pred_P^S

$H \xleftarrow{\$} \text{Fun}(\mathcal{K} \times \mathcal{D}, \mathcal{R}); \quad (x, z) \xleftarrow{\$} S(H); \quad x' \xleftarrow{\$} P(H, z)$
return $(x = x')$

Game Ext_D^S (secret seed)

$H \xleftarrow{\$} \text{Fun}(\mathcal{K} \times \mathcal{D}, \mathcal{R}); \quad (x, z) \xleftarrow{\$} S(H); \quad sd \xleftarrow{\$} \mathcal{K}$
 $y_0 \leftarrow H(sd, x); \quad y_1 \xleftarrow{\$} \mathcal{R}; \quad b \xleftarrow{\$} \{0, 1\}$
 $b' \xleftarrow{\$} D(H, y_b, z); \quad \textbf{return } (b = b')$

Figure 1: Prediction game (top) and the secret-seed extraction game (bottom). The first argument of S , P and D is the *full function table* of H ; the seed sd is drawn and used, but is withheld from D .

Definition 1. Set $\text{Adv}_{\mathcal{K}, \mathcal{D}, \mathcal{R}, S}^{\text{pred}}(P) := \Pr[\text{Pred}_P^S \Rightarrow 1]$ and

$$\text{Adv}_{\mathcal{K}, \mathcal{D}, \mathcal{R}}^{\text{ext}}(S, D) := 2 \Pr[\text{Ext}_D^S \Rightarrow 1] - 1.$$

A source S is ϵ -unpredictable if $\text{Adv}_{\mathcal{K}, \mathcal{D}, \mathcal{R}, S}^{\text{pred}}(P) \leq \epsilon$ for every unbounded predictor P .

Conjecture 1 (Secret seed). *There is a universal constant $c > 0$, independent of $\mathcal{K}$, $\mathcal{D}$, $\mathcal{R}$ and ϵ , such that for all nonempty finite $\mathcal{K}, \mathcal{D}, \mathcal{R}$, all $\epsilon \in (0, 1]$, every ϵ -unpredictable source S and every unbounded distinguisher D ,*

$$\text{Adv}_{\mathcal{K}, \mathcal{D}, \mathcal{R}}^{\text{ext}}(S, D) \leq \delta(\epsilon, K, D, R) := c \cdot \sqrt{\frac{\epsilon R + \log_2 D}{K}}.$$

The public-seed companion. This note states one conjecture, the one the accompanying page states. There is a companion statement in the same setting, differing only in that D is called as $D(H, sd, y_b, z)$ and does receive the seed; it conjectures the weaker bound $c(\sqrt{\epsilon R} + \sqrt{\log_2 D/K})$, in which the factor $1/K$ attaches to the second summand alone. That companion has been proved. It is a genuinely different claim, not a restatement: transposing the expression above verbatim into the public-seed game gives a *false* statement, which the companion's solution note establishes by counterexample. The two were originally posed together in a single write-up and were split so that each note states exactly the conjecture its page states.